

Quiz (Carolina Reaper)

- Q1.** What is the slope of a line perpendicular to $y = 3x + 2$?
(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 3
(D) -3
(E) 2
- Q2.** The equation of a line passing through $(1, 2)$ with a slope of -2 is $y =$
(A) $-2x + 4$
(B) $-2x - 4$
(C) $2x + 4$
(D) $2x - 4$
(E) $-2x + 0$
- Q3.** A truck is traveling on a road with a slope of $\frac{1}{4}$, what is the slope of a road perpendicular to it?
(A) -4
(B) 4
(C) $\frac{1}{4}$
(D) $-\frac{1}{4}$
(E) 2
- Q4.** The graph of $y = 2x - 1$ and $y = -\frac{1}{2}x + 1$ are
(A) parallel
(B) perpendicular
(C) neither parallel nor perpendicular
(D) the same line
(E) skew
- Q5.** A train is moving at an angle of 60° relative to the x -axis. What is the slope of its path?
(A) $\frac{1}{\sqrt{3}}$
(B) $\sqrt{3}$
(C) $-\frac{1}{\sqrt{3}}$
(D) $-\sqrt{3}$
(E) 1
- Q6.** Find the equation of a line perpendicular to $y = x + 2$ and passing through $(2, 3)$.
(A) $y = -x + 5$
(B) $y = -x + 1$
(C) $y = x + 1$
(D) $y = x - 1$
(E) $y = x - 5$
- Q7.** Two roads intersect at a 90° angle. If one road has a slope of 2, the slope of the other road is
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q8.** A bus is traveling along a road that makes an angle of 45° with the horizontal. The slope of this road is
(A) 1
(B) -1
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$
(E) $\frac{1}{\sqrt{2}}$

Open Ended Questions (FRQ)

- Q9.** A delivery truck is traveling from a warehouse to a store. The truck's path can be modeled by the equation $y = 2x - 1$. The store is located at $(4, 7)$. Find the equation of a line perpendicular to the truck's path that passes through the store.
- Q10.** The traffic flow on a highway can be modeled by two equations: $y = 3x + 2$ and $y = -\frac{1}{3}x + 1$. Find the point of intersection between the two lines and determine the angle between them. Show your work and explain your reasoning.

Chef's Tips

- Chef's Tips: Make sure students show all work and explain their reasoning.
- Use graph paper to help visualize the problems.
- Consider using real-world examples to illustrate the concepts.